Minimal Koldobsky Sets in ℓ_∞^n: Exact Face Factorization, Nonvertex Frames, and the Two-Dimensional Classification

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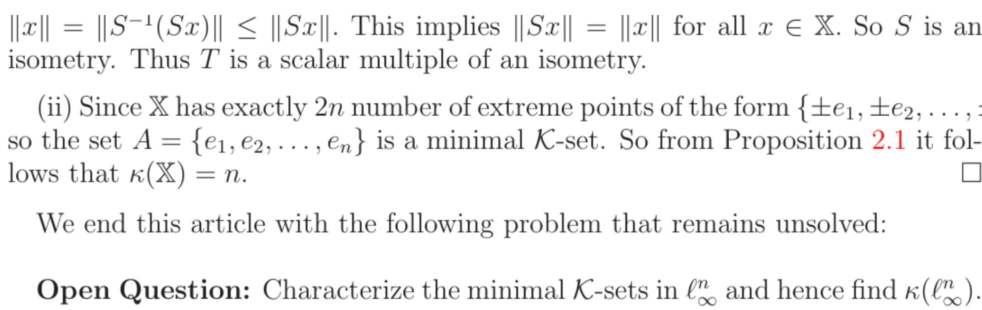
Status. Strong partial result, likely valid. We give an exact local criterion at every cube-face point, classify all finite minimal $\mathcal{K}$-sets in ℓ_∞^2, construct a continuum of nonvertex minimal n-point sets for every $n \geq 3$, classify all vertex-supported sets, and prove $\kappa(\ell_\infty^n) = n$. An intrinsic classification of all nonvertex minimal sets for $n \geq 3$ remains open.

1 Source question and results

Sain, Manna, and Paul introduce $\mathcal{K}$-sets through local preservation of Birkhoff–James orthogonality [1]. At the end of page 17 they ask:

Open Question. Characterize the minimal $\mathcal{K}$-sets in ℓ_∞^n and hence find $\kappa(\ell_\infty^n)$.


$\|x\| = \|S^{-1}(Sx)\| \leq \|Sx\|$. This implies $\|Sx\| = \|x\|$ for all $x \in \mathbb{X}$. So S is an isometry. Thus T is a scalar multiple of an isometry.

(ii) Since $\mathbb{X}$ has exactly $2n$ number of extreme points of the form $\{\pm e_1, \pm e_2, \ldots, \pm e_n\}$ so the set $A = \{e_1, e_2, \ldots, e_n\}$ is a minimal $\mathcal{K}$-set. So from Proposition 2.1 it follows that $\kappa(\mathbb{X}) = n$. $\square$

We end this article with the following problem that remains unsolved:

Open Question: Characterize the minimal $\mathcal{K}$-sets in ℓ_∞^n and hence find $\kappa(\ell_\infty^n)$.

Figure 1: The source question on page 17 of arXiv:2407.08900.

For a real normed space X, write

$$x \perp_B y \iff \|x + \lambda y\| \geq \|x\| \quad (\lambda \in \mathbb{R}).$$

A linear map $T : X \to X$ *preserves orthogonality at* x if $x \perp_B y$ implies $Tx \perp_B Ty$ for every y. A subset $A \subset S_X$ is a $\mathcal{K}$-set if every linear map preserving at all $x \in A$ is a scalar multiple of an isometry. It is minimal if no proper subset is a $\mathcal{K}$-set. The $\mathcal{K}$-number is the least cardinality of a finite $\mathcal{K}$-set.

We work in $X = \ell_\infty^n$. The packet proves four main theorems.

Theorem 1 (Arbitrary-face factorization)**.** *For $x \in S_{\ell_\infty^n}$ and $T \in \mathbb{L}(\mathbb{R}^n)$, suppose $w = Tx \neq 0$ and put $c = \|w\|_\infty$. Then T preserves orthogonality at x if and only if*

$$J(x) \subset c^{-1} T^* J(w), \tag{1}$$

where $J(u) = \{f \in S_{(\ell_\infty^n)^*} : f(u) = \|u\|_\infty\}$. Equivalently, the extreme norming functionals of x admit an entrywise nonnegative row-stochastic factorization through the normalized pullbacks of the extreme norming functionals of w. If $Tx = 0$, preservation at x is automatic.

Theorem 2 (Near-vertex nonvertex frames)**.** *Let $n \geq 3$ and choose arbitrary $t_1, \dots, t_n \in (-1, 1)$. Define*

$$x_j^i = \begin{cases} t_i, & j = i, \\ 1, & j \neq i. \end{cases}$$

Then $\{x^1, \dots, x^n\}$ is a minimal $\mathcal{K}$-set in ℓ_∞^n. In particular, every $n \geq 3$ has a continuum of genuinely nonvertex minimal $\mathcal{K}$-sets of optimal cardinality.

Theorem 3 (Complete finite classification in dimension two)**.** *Up to replacing points by antipodes, the finite minimal $\mathcal{K}$-sets in ℓ_∞^2 are exactly*

(i) *two linearly independent cube vertices;*

(ii) *one cube vertex, one smooth boundary point whose first coordinate is active, and one smooth boundary point whose second coordinate is active.*

Theorem 4 (Vertex sets and the exact $\mathcal{K}$-number)**.** *Let $\mathcal{E}_n = \{-1, 1\}^n$ and $A \subset \mathcal{E}_n$. Then*

$$A \text{ is a } \mathcal{K}\text{-set} \quad \Longleftrightarrow \quad \operatorname{span} A = \mathbb{R}^n.$$

Thus the minimal vertex-supported sets are exactly vertex bases, and for every $n \geq 2$,

$$\boxed{\kappa(\ell_\infty^n) = n}.$$

The restriction "finite" in Theorem 3 is intentional. No claim about possible infinite minimal sets is made.

2 Idea of the proof

At an arbitrary boundary point, Birkhoff–James orthogonality is a straddling-zero condition on the finite list of active signed coordinate functionals. If the normalized pullback list at the image is strictly positive on a direction, preservation forces the source list to have one strict sign. A segment from the distinguished point rules out the negative sign. Passing to the closed cone and applying Farkas duality produces the exact stochastic factorization of norming simplices.

At a cube vertex the source simplex spans the whole dual space, so the local factorization forces an invertible operator, a vertex image, and a sign-conjugated nonnegative inverse. With a vertex normalized to $\mathbf{1}$, the inverse becomes row-stochastic. Every additional test point then says exactly that it lies in the stochastic image of the cube.

The nonvertex construction takes n points just inside the n different coordinate directions from one vertex. Their norming simplices overlap in $n-2$ coordinates. These overlaps first exclude singular preservers and then equalize all image norms and all inverse-row ℓ_1 norms. Equality in Hölder's inequality fixes signs on row supports. Any shared support coordinate becomes constant across all image columns, while each inverse row needs a private coordinate because the corresponding diagonal parameter is strictly below one. There are exactly enough coordinates for one private coordinate per row, forcing a monomial inverse.

In dimension two, a normalized vertex makes the inverse stochastic, and one smooth test of each active-coordinate type makes one row at a time deterministic. This proves the three-point set; diagonal and triangular stochastic deformations show that every missing finite type fails. For vertex bases in all dimensions, the positive inverse laws match source and image row-sign patterns and again force a monomial operator.

3 The exact arbitrary-face certificate

For $u \in S_{\ell_\infty^n}$ let

$$I(u) = \{i : |u_i| = 1\}.$$

Let F_u be the $|I(u)| \times n$ matrix whose row indexed by $i \in I(u)$ is $\operatorname{sgn}(u_i)e_i^*$. The norming set is the simplex

$$J(u) = \operatorname{co}\{\operatorname{sgn}(u_i)e_i^* : i \in I(u)\}. \tag{2}$$

James's criterion [2] gives

$$u \perp_B z \quad \Longleftrightarrow \quad \min(F_u z) \le 0 \le \max(F_u z). \tag{3}$$

Proof of Theorem 1. Only the case $w = Tx \neq 0$ needs proof. Set

$$A = F_x, \qquad B = c^{-1} F_w T.$$

Every row of either matrix evaluates to one at x, so $Ax = Bx = \mathbf{1}$. By (3), preservation says that if the coordinates of Az straddle zero, then those of Bz straddle zero.

We first claim

$$Bz > 0 \quad \Longrightarrow \quad Az > 0. \tag{4}$$

Indeed, the contrapositive of preservation says that Az is either strictly positive or strictly negative. In the latter case consider $z(t) = (1-t)x + tz$. The vector $Bz(t)$ remains strictly positive for $0 \le t \le 1$, whereas $Az(t)$ moves from $\mathbf{1}$ to a strictly negative vector and hence meets the straddling set. This contradicts preservation and proves (4). Applying it to $z + \varepsilon x$ and letting $\varepsilon \downarrow 0$ yields

$$Bz \ge 0 \quad \Longrightarrow \quad Az \ge 0. \tag{5}$$

For each row a of A, implication (5) says that a is nonnegative on the polyhedral cone $\{z : Bz \ge 0\}$. The Farkas lemma identifies the dual cone with the nonnegative cone generated by the rows of B. Hence $A = CB$ for an entrywise nonnegative matrix C. Evaluating at x gives $\mathbf{1} = C\mathbf{1}$, so C is row-stochastic. Using (2), this is exactly (1).

Conversely, if $A = CB$ for a nonnegative row-stochastic C, then $Bz > 0$ implies $Az > 0$, and $Bz < 0$ implies $Az < 0$. The contrapositive is precisely the required straddling implication. Finally, if $Tx = 0$, the conclusion $Tx \perp_B Ty$ holds for every y by definition. □

Remark 5. Theorem 1 turns local preservation into a finite linear feasibility certificate. It is exact, rather than merely a comparison of the dimensions of the source and image faces.

4 Vertices and stochastic parallelotopes

For a cube vertex x, put $D_x = \operatorname{diag}(x_1, \ldots, x_n)$.

Corollary 6 (Local positive-inverse law)**.** *Let $x \in \mathcal{E}_n$ and $Tx \neq 0$. Then T preserves at x if and only if there are $c > 0$ and $v \in \mathcal{E}_n$ such that*

$$Tx = cv, \qquad T \text{ is invertible}, \qquad cD_xT^{-1}D_v \geq 0$$

entrywise.

Proof. The simplex $J(x)$ contains n linearly spanning extreme points. Inclusion (1) forces T^* to be surjective and the image face to have at least n active coordinates. Thus T is invertible and every coordinate of Tx is active, so $Tx = cv$ for a vertex v. In the matrix factorization from Theorem 1,

$$D_x = C(c^{-1}D_vT),$$

whence $C = cD_xT^{-1}D_v$. Necessity and sufficiency now follow from the factorization theorem. □

The vertex law has a useful global reformulation. Suppose a test set contains a vertex. Source and output cube isometries and scalar normalization reduce the nonzero case to

$$T\mathbf{1} = \mathbf{1}, \qquad R = T^{-1} \geq 0, \qquad R\mathbf{1} = \mathbf{1}. \tag{6}$$

Thus R is invertible and row-stochastic.

Proposition 7 (Stochastic-parallelotope criterion)**.** *Under (6), for every $x \in S_{\ell_\infty^n}$,*

$$T \text{ preserves at } x \iff \|R^{-1}x\|_\infty = 1 \iff x \in R(B_{\ell_\infty^n}).$$

Proof. Put $y = R^{-1}x = Tx$. Since R is row-stochastic, $\|Ry\|_\infty \leq \|y\|_\infty$; hence $\|x\|_\infty = 1$ shows that $\|y\|_\infty \leq 1$ is equivalent to $\|y\|_\infty = 1$.

Assume $\|y\|_\infty = 1$. If $i \in I(x)$, equality in

$$x_i = \sum_j R_{ij}y_j$$

forces $y_j = \operatorname{sgn}(x_i)$ wherever $R_{ij} > 0$. Consequently $g_i = \operatorname{sgn}(x_i)\sum_j R_{ij}e_j^*$ belongs to $J(y)$, and $T^*g_i = \operatorname{sgn}(x_i)e_i^*$. Theorem 1 gives preservation. Conversely, the same theorem represents every extreme functional $\operatorname{sgn}(x_i)e_i^*$ as $c^{-1}T^*g_i$ with $g_i \in J(y)$ and $c = \|y\|_\infty$. Thus $g_i = cR^T \operatorname{sgn}(x_i)e_i^*$; its ℓ_1 norm is c because row i of R has sum one. Since g_i is norming, its norm is one, so $c = 1$. □

Thus every vertex-containing instance reduces exactly to asking which test sets force a stochastic image parallelotope $R(B_\infty) \subset B_\infty$ that contains them to arise from a permutation matrix.

5 A continuum of nonvertex minimal sets

Proof of Theorem 2. Let $X = (x^1 \ \cdots \ x^n)$ and $D = \operatorname{diag}(1 - t_1, \ldots, 1 - t_n)$. Then $X = \mathbf{1}\mathbf{1}^T - D$. The matrix determinant lemma gives

$$\det X = (-1)^n \prod_i (1 - t_i) \left(1 - \sum_i \frac{1}{1 - t_i}\right). \tag{7}$$

Every summand is greater than $1/2$, so the parenthesis is nonzero for $n \geq 3$. Hence the x^i form a basis.

Let T preserve at every x^i. Suppose first that T is singular but $Tx^i \neq 0$ for some i. The simplex inclusion in Theorem 1 puts

$$H_i = \operatorname{span}\{e_k^* : k \neq i\}$$

inside $\operatorname{range} T^*$. Therefore $\operatorname{rank} T = n-1$ and the range equals H_i. If $Tx^j \neq 0$ for a second index j, the same argument gives $H_i = H_j$, impossible. Thus at most one basis image is nonzero and $\operatorname{rank} T \leq 1$, contradicting $n-1 \geq 2$. If all images vanish then $T = 0$. We may therefore assume that a nonzero T is invertible.

Write $R = T^{-1}$, let r_k be its kth row, and set

$$\rho_k = \|r_k\|_1, \qquad c_i = \|Tx^i\|_\infty.$$

For $k \neq i$, e_k^* is an extreme norming functional of x^i. Invertibility and Theorem 1 say that its unique pulled-back norming functional is $c_i R^T e_k^*$. Taking its dual norm gives

$$c_i \rho_k = 1 \qquad (k \neq i). \tag{8}$$

For any two indices there is a third; the overlaps in (8) show that all c_i equal a common c and all ρ_k equal c^{-1}. Rescale T so that both common values are one.

Put $W = TX = (w^1 \ \cdots \ w^n)$. Then

$$RW = X, \qquad \|w^i\|_\infty = 1, \qquad \|r_k\|_1 = 1.$$

For $k \neq i$, the identity $r_k \cdot w^i = 1$ is equality in Hölder's inequality. Hence

$$r_{kj} \neq 0, k \neq i \quad \Longrightarrow \quad w_j^i = \operatorname{sgn}(r_{kj}). \tag{9}$$

Suppose output coordinate j belongs to the supports of two different rows r_k, r_ℓ. Choose i different from both. Equation (9) first shows that the two signs agree. Applying the same equation with row k to every column other than k, and with row ℓ to column k, shows that the entire jth row of W is constant with that sign.

Each row r_k must now have a *private* support coordinate, one which belongs to no other row. Otherwise every coordinate in its support is shared, so the preceding observation gives

$$(RW)_{kk} = r_k \cdot w^k = \|r_k\|_1 = 1,$$

contrary to $(RW)_{kk} = X_{kk} = t_k < 1$. Private coordinates for distinct rows are distinct. There are n rows and only n coordinates, so each row has exactly one support coordinate and no coordinate is shared. Thus R is monomial. Since all row ℓ_1 norms equal one, R is a signed permutation. Undoing the normalization, T is a scalar multiple of an isometry.

Finally, every $\mathcal{K}$-set must span: if it does not, a nonzero rank-one map annihilating its span preserves there and is not a scalar isometry. Since our set is an n-point basis, removal of any point destroys the spanning property. It is minimal. $\square$

6 Complete finite classification in the square

Replacing x by $-x$ does not change local preservation. Projectively, the square boundary consists of vertices, smooth points active in coordinate one, and smooth points active in coordinate two.

Proof of Theorem 3. First consider a triple of the stated form. Normalize its vertex to $\mathbf{1} = (1, 1)$. If $T\mathbf{1} = 0$, write the columns of T as $a, -a$. A smooth active-first point may be written $(1, u)$ with $|u| < 1$ and satisfies $(1, u) \perp_B e_2$. Preservation requires $(1 - u)a \perp_B -a$, which forces $a = 0$. Thus $T = 0$.

If $T\mathbf{1} \neq 0$, normalize as in (6), so $R = T^{-1}$ is invertible and row-stochastic. At a smooth point active in coordinate i, Proposition 7 gives $x = Ry$ for some $y \in B_\infty$. Equality in active row i forces y to have the active sign on the support of that row. If both entries of the row were positive, y would be a constant vertex; every stochastic row would then give the same constant vertex x, contradicting smoothness. Hence row i is a coordinate row. One smooth point of each active type forces both rows of R to be distinct coordinate rows. Therefore R is a permutation and T is a scalar cube isometry. The triple is a $\mathcal{K}$-set.

We next show that the list is exhaustive and minimal. A finite set of smooth points only is preserved by

$$T = \operatorname{diag}(1, r)$$

for some $r > 1$ sufficiently close to one: all active-first points retain first-coordinate activity, and all active-second points retain second-coordinate activity. This T is not a scalar isometry.

After normalizing a vertex to $\mathbf{1}$, a finite collection of smooth points of active type one may be written projectively as $(1, u_s)$, $|u_s| < 1$. Choose

$$0 < a < \min_s \frac{1 + u_s}{2}, \qquad R = \begin{pmatrix} 1 & 0 \\ a & 1 - a \end{pmatrix}.$$

Then

$$R^{-1}(1, u_s)^T = \left(1, \frac{u_s - a}{1 - a}\right)^T \in B_\infty.$$

Proposition 7 says that the nonisometry R^{-1} preserves at the vertex and at all these points. The transposed construction handles active type two.

Finally, two nonantipodal projective vertex directions are linearly independent and hence form a $\mathcal{K}$-set by Theorem 4, whose proof appears below. A finite minimal set containing two such directions is therefore exactly that pair. A minimal set with only one vertex direction must contain both smooth types, while any one point of each type with the vertex already forms the triple proved sufficient above. Antipodal repeats are redundant. These observations prove both exhaustiveness and minimality. $\square$

7 Vertex-supported classification and the $\mathcal{K}$-number

We include the vertex proof because it both supplies Theorem 3 and determines the numerical invariant in the source question.

Lemma 8 (Common image norm)**.** *If $x, y \in \mathcal{E}_n$ and T preserves at both points, then $\|Tx\|_\infty = \|Ty\|_\infty$.*

Proof. The conclusion is immediate for $y = \pm x$. Otherwise, x and y agree in some coordinate and disagree in another, so the vertex straddling criterion gives $x \perp_B (x + y)$ and $y \perp_B (x + y)$. If one image vanishes, preservation forces the other to be orthogonal to itself and hence to vanish. Otherwise Corollary 6 writes $Tx = cv$ and $Ty = dw$ for $c, d > 0$ and vertices v, w. Preservation gives

$$cv \perp_B (cv + dw), \qquad dw \perp_B (cv + dw).$$

Choosing scalar -1 in the definition of orthogonality in the first relation yields $d \geq c$; the second yields $c \geq d$. $\square$

Proof of Theorem 4. If A does not span, choose a nonzero functional f annihilating its span and a nonzero vector u. The rank-one map $Tz = f(z)u$ vanishes on A, hence preserves there, but is not a scalar isometry for $n \geq 2$.

Conversely, choose a vertex basis $x^1, \dots, x^n$ from a spanning A and suppose T preserves there. By Lemma 8, their image norms have a common value c. If $c = 0$, then $T = 0$. Otherwise Corollary 6 gives vertices v^i with $Tx^i = cv^i$, makes T invertible, and gives

$$cD_{x^i}T^{-1}D_{v^i} \geq 0 \qquad (i = 1, \dots, n). \tag{10}$$

Let $X = (x^1 \ \cdots \ x^n)$, $V = (v^1 \ \cdots \ v^n)$, and $R = T^{-1} = (r_{ab})$. Both X and $V = c^{-1}TX$ are invertible. If $r_{ab} \neq 0$, inequalities (10) imply

$$(v_b^1, \dots, v_b^n) = \operatorname{sgn}(r_{ab})(x_a^1, \dots, x_a^n). \tag{11}$$

Two nonzero entries in column b of R would make two rows of X proportional, contrary to invertibility. Every column therefore contains exactly one nonzero entry, and invertibility puts them in distinct rows. Thus R and T are monomial. Since $Tx^i = cv^i$, every nonzero monomial entry of T has absolute value c. Hence T is a scalar signed permutation. This proves the equivalence, and inclusion-minimal spanning vertex sets are precisely bases.

The same rank-one annihilator shows that every finite $\mathcal{K}$-set, not only a vertex set, has at least n points. Vertex bases exist; for example, the columns defined by $x_j^i = -1$ for $j < i$ and $x_j^i = 1$ for $j \geq i$ form a basis. The lower bound is attained, proving $\kappa(\ell_\infty^n) = n$. $\square$

8 A sharp obstruction to the naive coordinate test

One might hope that $\{\mathbf{1}, e_1, \dots, e_n\}$ is a $\mathcal{K}$-set. Already for $n = 3$ this fails. Let J be the all-ones matrix and put

$$T = J - 2I, \qquad T^{-1} = \tfrac{1}{2}(J - I). \tag{12}$$

The inverse is nonnegative and row-stochastic, and $T\mathbf{1} = \mathbf{1}$, so Corollary 6 gives preservation at $\mathbf{1}$. Moreover, Te_i has -1 in coordinate i and 1 in the other two coordinates, and

$$g_i = \tfrac{1}{2}\sum_{j \neq i} e_j^* \in J(Te_i), \qquad T^* g_i = e_i^*.$$

Theorem 1 gives preservation at every e_i. But T is not a scalar signed permutation. Taking the block sum of this 3×3 matrix with an identity gives the same obstruction in all $n > 3$.

This counterexample explains why Theorem 2 uses points with $n - 1$ overlapping active coordinates: one smooth coordinate test in each direction does not give enough rigidity.

9 Verification, literature boundary, and limitations

The script

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code/verify_cube_vertex_k_sets.py
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implements Theorem 1 as a linear feasibility test. It retains the exhaustive vertex-image enumeration through dimension four, the 8,800-point positive-inverse grid check, and 1,000 random vertex-basis checks. It also exhausts all $3^9 = 19{,}683$ ternary matrices on the near-vertex frame

$$(0,1,1), \quad (1,0,1), \quad (1,1,0),$$

finding precisely the 48 nonzero signed permutations; checks 200 random near-vertex frames through dimension seven; exhausts 625 small integer matrices on a representative two-dimensional triple and checks nonisometries for its proper pair types; and verifies (12). It terminates with

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ALL UNRESTRICTED K-SET PROGRAM CHECKS PASSED.
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These are stress tests, not substitutes for the analytic proofs.

Bounded searches on August 9, 2026 used the run indexes and arXiv with “minimal K-sets ell-infinity”, “Koldobsky-set”, “K-number orthogonality isometries”, and close variants. The source [1] and the later broad refinement [3] were inspected. No exact statement of the arbitrary-face factorization, near-vertex theorem, dimension-two classification, vertex-spanning theorem, or exact $\mathcal{K}$-number was found. This is a bounded novelty assessment, not an exhaustive review.

The factorization theorem is a complete local certificate and reduces every finite candidate to explicit nonnegative matrix constraints. It has not yet been converted into a closed intrinsic classification of all finite nonvertex minimal sets for $n \geq 3$. Possible infinite minimal sets are also outside the claim. These are the precise remaining boundaries.

Human review should focus on the Farkas step in Theorem 1, the singular case and private-coordinate argument in Theorem 2, and the finite-set qualifier in Theorem 3. Recommended status: *strong partial result, likely valid*.

References

[1] D. Sain, J. Manna, and K. Paul, *On local preservation of orthogonality and its application to isometries*, Linear Algebra Appl. 690 (2024), 112–131; arXiv:2407.08900.

[2] R. C. James, *Orthogonality and linear functionals in normed linear spaces*, Trans. Amer. Math. Soc. 61 (1947), 265–292.

[3] J. Manna, K. Mandal, K. Paul, and D. Sain, *Refinements of the Blanco–Koldobsky–Turnšek Theorem*, Electron. J. Linear Algebra 41 (2025), 616–631; arXiv:2504.11161.